



Lexan* Resin 945

Americas: COMMERCIAL

Lexan* 945 Polycarbonate (PC) resin is a non-filled, injection moldable grade. This non-chlorinated, non-brominated flame retardant PC has an UL-94 V0 rating and is available in various opaque color options. Lexan 945 is a general purpose resin designed to meet the needs of various applications.

TYPICAL PROPERTIES ¹	TYPICAL VALUE	Unit	Standard
MECHANICAL			
Tensile Stress, yld, Type I, 2.0 in/min	8900	psi	ASTM D 638
Tensile Stress, brk, Type I, 2.0 in/min	9400	psi	ASTM D 638
Tensile Strain, yld, Type I, 2.0 in/min	6	%	ASTM D 638
Tensile Strain, brk, Type I, 2.0 in/min	125	%	ASTM D 638
Tensile Modulus, 2.0 in/min	329000	psi	ASTM D 638
Flexural Stress, yld, 0.05 in/min, 2 in span	14600	psi	ASTM D 790
Flexural Modulus, 0.05 in/min, 2 in span	339000	psi	ASTM D 790
IMPACT			
Izod Impact, notched, 73°F	15	ft-lb/in	ASTM D 256
Instrumented Impact Total Energy, 73°F	649	in-lb	ASTM D 3763
THERMAL			
Vicat Softening Temp, Rate B/50	290	°F	ASTM D 1525
HDT, 66 psi, 0.125 in, unannealed	280	°F	ASTM D 648
HDT, 264 psi, 0.125 in, unannealed	260	°F	ASTM D 648
CTE, flow, -40°F to 100°F	3.7E-05	1/°F	ASTM E 831
CTE, xflow, -40°F to 100°F	3.7E-05	1/°F	ASTM E 831
Relative Temp Index, Elec	130	°C	UL 746B
Relative Temp Index, Mech w/impact	120	°C	UL 746B
Relative Temp Index, Mech w/o impact	130	°C	UL 746B
PHYSICAL			
Specific Gravity	1.19	-	ASTM D 792
Mold Shrinkage, flow, 0.125" (5)	0.6 - 0.8	%	SABIC Method
Melt Flow Rate, 300°C/1.2 kgf	10	g/10 min	ASTM D 1238
ELECTRICAL			
Arc Resistance, Tungsten {PLC}	7	PLC Code	ASTM D 495

(1) Typical values only. Variations within normal tolerances are possible for various colors. All values are measured after at least 48 hours storage at 23±176;C/50% relative humidity. All properties, except the melt volume and melt flow rates, are measured on injection molded samples. All samples tested under ISO test standards are prepared according to ISO 294.

(2) Only typical data for selection purposes. Not to be used for part or tool design.
 (3) This rating is not intended to reflect hazards presented by this or any other material under actual fire conditions.
 (4) Internal measurements according to UL standards.
 (5) Measurements made from laboratory test coupon. Actual shrinkage may vary outside of range due to differences in processing conditions, equipment, part geometry and tool design. It is recommended that mold shrinkage studies be performed with surrogate or legacy tooling prior to cutting tools for new molded article.

Source GMD, last updated:

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TYPICAL PROPERTIES ¹	TYPICAL VALUE	Unit	Standard
ELECTRICAL			
Hot Wire Ignition {PLC}	2	PLC Code	UL 746A
High Voltage Arc Track Rate {PLC}	4	PLC Code	UL 746A
High Ampere Arc Ign, surface {PLC}	3	PLC Code	UL 746A
Comparative Tracking Index (UL) {PLC}	2	PLC Code	UL 746A
FLAME CHARACTERISTICS			
UL Recognized, 94V-2 Flame Class Rating (3)	0.031	in	UL 94
UL Recognized, 94V-0 Flame Class Rating (3)	0.045	in	UL 94
UL Recognized, 94-5VA Rating (3)	0.12	in	UL 94
Glow Wire Flammability Index 960°C, passes at	0.039	in	IEC 60695-2-12
Glow Wire Ignitability Temperature, 1.0 mm	875	°C	IEC 60695-2-13
Oxygen Index (LOI)	35	%	ISO 4589

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PROCESSING PARAMETERS	TYPICAL VALUE	Unit
Injection Molding		
Drying Temperature	250	°F
Drying Time	3 - 4	hrs
Drying Time (Cumulative)	48	hrs
Maximum Moisture Content	0.02	%
Melt Temperature	560 - 600	°F
Nozzle Temperature	550 - 590	°F
Front - Zone 3 Temperature	560 - 600	°F
Middle - Zone 2 Temperature	540 - 580	°F
Rear - Zone 1 Temperature	520 - 560	°F
Mold Temperature	160 - 200	°F
Back Pressure	50 - 100	psi
Screw Speed	40 - 70	rpm
Shot to Cylinder Size	40 - 60	%
Vent Depth	0.001 - 0.003	in

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